ACADEMIC STRESS HEATMAP WITH PERSONALIZED FINGERPRINT OVERLAY

25-26J-480

Project Proposal Report

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

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Prof. Samantha Thelijjagoda

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Signature of the Co-Supervisor Date

Ms. Chamali Pabasara

Abstract

Academic stress is increasingly recognized as one of the most significant challenges faced by university students worldwide. The demands of coursework, examinations, and extracurricular activities often create high-pressure environments, leading to anxiety, depression, and poor academic performance. While traditional counseling and wellness services are valuable, they often have limitations regarding accessibility, stigma, and the ability to provide timely and personalized interventions. This project proposes the development of a Self-Monitoring and Anonymous Counseling-Based Digital Companion for Academic Stress Management, which is a mobile platform designed to equip students with tools to monitor, predict, and manage stress. The system will integrate four key components: mood and journaling-based self-monitoring, anonymous counseling modules, a novel adaptive stress forecasting and visualization system (the central innovation of this project), and a personalized recommendation engine. The Adaptive Academic Stress Heatmap with Personalized Fingerprint Overlay is the most significant contribution of this research. By combining self-reported data (such as mood logs and journals), academic workload and deadlines, and passive IoT data (including sleep quality, screen time, and physical activity), the system will generate unique stress fingerprints for each student and forecast upcoming high-stress periods three to five days in advance. These insights will be presented through an interactive heatmap calendar, paired with personalized, context-aware coping strategies. The proposed solution has the potential to help students gain self-awareness, anticipate stress before it escalates, and adopt healthier coping mechanisms. By addressing critical gaps in personalization, predictive capability, and the integration of academic context, this project contributes both to the field of mental health informatics and to the broader goal of improving student well-being.

Keywords *: Stress Fingerprint, Academic Heatmap, Machine Learning*

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LIST OF ABBREVIATIONS

Abbreviation	Description
ML	Machine Learning
AI	Artificial Intelligence
PSS	Perceived Stress Scale
IoT	Internet of Things

1. INTRODUCTION

Stress is an overwhelming sense of worry and mental strain that arises when facing challenges. It is a core aspect of human experience, igniting our instincts to confront and navigate obstacles and threats that life presents. Everyone engages with stress in their unique way, experiencing its weight and intensity in varied degrees as they traverse the complexities of their daily lives [1]. In recent years, the issue of academic stress among young individuals, particularly those aged 16 to 30, has become a significant and growing concern. Factors such as heavy academic workload, tight deadlines, examination anxiety, social expectations, and poor coping mechanisms have all contributed to an increase in emotional burnout, anxiety, and even depression. A study conducted by the Franciscan University Counseling Center indicated that the top three concerns regarding mental health among university students were academic performance, pressure to succeed, and post-graduation plans [2]. Many students find it challenging to manage the increasing demands of their education, often moving to negative coping strategies and harmful behaviors. These responses can include substance abuse, social withdrawal, self-harm, and, in extreme circumstances, suicidal intentions. According to the World Health Organization, half of the mental health disorders in adulthood start at the age of 18, but most cases are undetected and untreated [3].

In response, digital mental health tools have emerged as a scalable solution to this issue. However, generic wellness apps typically provide generic advice and do not adapt to individual patterns [4]. Also, recent studies have highlighted the potential benefits of personalized multidimensional monitoring, which integrates physiological sensors with contextual measures, in enhancing stress awareness. [5].

Our proposed project aims to build on these insights. We plan to develop a personalized mobile application that visualizes each student's stress levels over time and gives customized coping suggestions.

This application features several components designed to support student stress monitoring and management. These include:

1. **A Speech-Based Reflective Support Module for monitoring student stress.**

   This feature will allow students to express their feelings verbally, offering a seamless way to track and monitor their stress in real-time.

2. **An Emotionally Intelligent Personal Assistant for real-time stress detection and adaptive therapeutic support.**

   This cutting-edge assistant will enable real-time stress detection, employing compassionate algorithms to provide adaptive and therapeutic support precisely when it's needed most.

3. **Daily monitoring of emotional states that provides real-time coping strategies.**

   Students will benefit from regular check-ins on their emotional states, equipping them with immediate coping strategies to address their feelings as they arise.

4. **An Adaptive Academic Stress Heatmap with a Personalized Fingerprint Overlay.**

   This unique feature will overlay personalized stress patterns onto an academic timeline, visualizing correlations between academic pressures and stress responses, tailored through individual data.

Our particular focus in this proposal will be the development of the academic stress heatmap component. To construct this intricate tool, we will gather rich data concerning academic life, including critical deadlines, upcoming exams, and personal mood logs and journals. Through this data, we will extract meaningful behavioral features, segment users based on their unique stress response patterns, anticipate future stress levels, and adaptively visualize these insights to empower students. The following sections survey the literature on stress detection, identify gaps, and outline our proposed methods.

1.1. Background and Literature Survey

Academic stress is a significant challenge for university students [6]. Stress can be triggered by various factors, including heavy workloads and exams, as well as contextual and behavioral elements like poor time management, irregular sleep patterns, and personal coping styles [7]. University settings have unique stressors, such as cyclical events like exams, project deadlines, and assignments, which lead to predictable yet personally significant stress peaks. Importantly, students respond to stress in different ways: some experience heightened anxiety as deadlines approach, others are significantly affected by inadequate sleep, and some struggle primarily with the accumulation of workload. Traditional stress management tools can be helpful, but they often use general strategies that do not address these individual patterns. A digital system that can recognize these differences by detecting and predicting stress based on unique behavioral patterns can provide more meaningful and proactive support tailored to the needs of students.

Recent research confirms that university students face ever-increasing stress, making real-time stress monitoring and predictions crucial more than ever [8]. Studies showed that smartphones and wearable sensor data can predict stress levels and academic engagement with significant accuracy [9]. This opens opportunities to shift from reactive stress management to predictive, personalized support systems. However, the effectiveness of digital stress tools hinges on personalization. A review of 150 stress-management apps found personalization to be the most common strategy, followed by self-monitoring [10].

StudentLife [11] is implemented as a university-scale smartphone sensing system using Android phones that captures mobility, sleep activity, mood logs, and academic performances using a single class of 48 students across a 10-week term at Dartmouth College. It outlined several significant connections between the automatic objective sensor data from smartphones and mental health and educational outcomes of the student body. Even though these results were useful for research purposes and

provided strong evidence that passive sensing relates to academic stress, this system lacked personalized forecasting and was not deployed as a tool for students.

Another research on mental stress detection using wearable sensors [12] has investigated different stress detection approaches that can be adopted with wearable devices with sensors, including Electrocardiogram (ECG), Electroencephalography (EEG), and Photoplethysmography (PPG). The research highlights the importance of context by examining stress levels in different environments, such as while driving, studying, and working, where stress responses can vary significantly. Additionally, the study has introduced an innovative multimodal stress detection system that leverages deep learning techniques based on data collected from these wearable sensors. This approach aims to enhance the accuracy and reliability of stress detection processes by combining multiple physiological signals, providing a comprehensive understanding of an individual's stress levels in real-time.

A team of researchers from Japan has created a mobile application called NokiriMe, which allows students to measure and track their stress over time [13]. This app enables users to not only measure their stress but also to visualize patterns and identify correlations between their stress and various psychological responses. One of the key features of NokiriMe is an academic stress questionnaire, which helps users identify their current stress levels. However, feedback from users indicates that the app could significantly improve in terms of usability and user engagement. To enhance the overall experience, many users have suggested incorporating motivational messages that could inspire and uplift them during challenging times. Additionally, they propose the inclusion of personalized coping strategies tailored to individual stress profiles, which would help students better manage their stress effectively.

Personalization in stress modeling is still an area that requires further exploration. One study has implemented a multitask deep learning framework, which led to a significant improvement in stress state classification by identifying individual patterns based on sensor data [14]. This finding demonstrates the potential for generating personalized predictions. The idea of "stress fingerprinting" builds on this approach. Instead of applying the same thresholds for stress detection across all individuals, students can

be grouped into clusters based on their responses to various triggers. This method allows the system to adjust its sensitivity and recommendations according to whether a student is more affected by workload, sleep irregularities, or other factors. However, only a few existing systems have successfully implemented this clustering method in the context of academic stress management.

The use of visual representation is crucial for improving user engagement and enhancing awareness. Specifically, calendar-style heatmaps serve as an effective tool for representing behavioral patterns over time, making them particularly useful for applications like task planning and mood tracking. However, there is a significant gap in systems that integrate forecasted stress levels within students' academic calendars. By creating a clear overlay that maps predicted stress levels on a color-coded spectrum from green hues representing low stress to intense reds indicating high stress, students can gain immediate insights into their upcoming academic demands. This feature enables them to pinpoint potential periods of high stress in advance, allowing for planning and the development of tailored coping strategies.

1.2. Research Gap

While many digital mental health tools are available for the general population, there is a significant research gap regarding stress, specifically among university students. This gap particularly concerns personalized, predictive, and visual interventions that are directly linked to academic workload. Most mobile mental health apps do not address specific stressors, such as assignment deadlines, exam schedules, or semester workload patterns, which are crucial for students. Also, some findings state that some apps do not respect privacy and are not designed with users in mind [15]. Additionally, it is stated that there is a lack of clinically validated evidence of the efficacy of mental health applications [16]. Despite advances in machine learning, few apps utilize predictive algorithms to anticipate stress levels days in advance, especially based on time-series data such as mood logs, sleep patterns, and workload. The proposed system aims to incorporate multimodal time-series forecasting to predict stress levels 3 to 5 days in advance, thereby addressing this predictive intelligence gap. Visualization also remains unaddressed. Stress management apps typically present data through simple charts or progress trackers. It is missing a dynamic, calendar-based heatmap that overlays academic tasks with predicted stress zones, giving students an intuitive and immediate understanding of how their workload, habits, and behaviors may converge to create high-stress periods.

This proposal seeks to close these gaps by developing a system that combines predictive analytics, academic workload integration, and personalized visualization into a cohesive tool.

Capability / System	[11]	[12]	[13]	[17]	[18]	[19]	**Our Solution**
Uses physiological / IoT data	✓	✓	✓	✓	×	×	✓
Questionnaire to detect stress	×	×	✓	×	×	×	✓
Personalized stress fingerprint via clustering	×	×	×	×	×	×	✓

Academic Stress Heatmap	×	×	×	×	×	×	✓
Mood and journaling input integration	×	×	×	×	×	×	✓
Emotionally intelligent personal assistant	×	×	×	×	✓	✓	✓

Table 1 : Comparative analysis with existing solutions

1.3. Research Problem

The primary research problem is how to effectively monitor, predict, and manage academic stress among university students using a personal digital companion. This companion will integrate multimodal data, machine learning, and academic-context visualization, all while maintaining anonymity and accessibility.

The research area for the academic heatmap and the personalized stress fingerprint lies at the intersection of mental health informatics and personalized digital inventions. Stress is not uniform; instead, it varies by individual depending on factors such as workload, sleep quality, and lifestyle habits. Also, wearable and smartphone-based passive sensing (e.g., sleep tracking, screen time) provides rich behavioral data that can complement self-reported measures such as mood logs and journals. These data streams, when processed with AI and predictive modeling, allow the construction of personalized stress profiles and the anticipation of high-stress events before they escalate. Using these captured data, users need to be categorized according to their behavioral responses to stress, like being deadline-sensitive, sleep-reactive, or high-load avoidant. These labels reflect real, observable differences in how people experience and cope with stress. Based on the user's category, coping strategies should be suggested to reduce potential stress occurring in the next 3 to 5 days.

2. OBJECTIVES

2.1. Main Objectives

This project aims to design and implement a digital companion for academic stress management that integrates self-monitoring, anonymous counseling, predictive visualization, and adaptive coping strategies.

This component aims to design and implement a stress management tool that provides personalized and predictive support through stress fingerprinting and visualization.

2.2. Specific Objectives

- To collect and integrate active (onboarding questionnaire, mood logs) and passive (screen-time, sleep-time) data with academic deadlines.
- To model stress fingerprints by categorizing users into distinct clusters based on their unique response types and stress patterns.
- To develop a predictive model that is capable of forecasting potential stress periods 3 to 5 days in advance.
- To implement an intuitive heatmap visualization that overlays important academic deadlines with the projected stress periods.

3. METHODOLOGY

3.1. System Architecture and Overall View

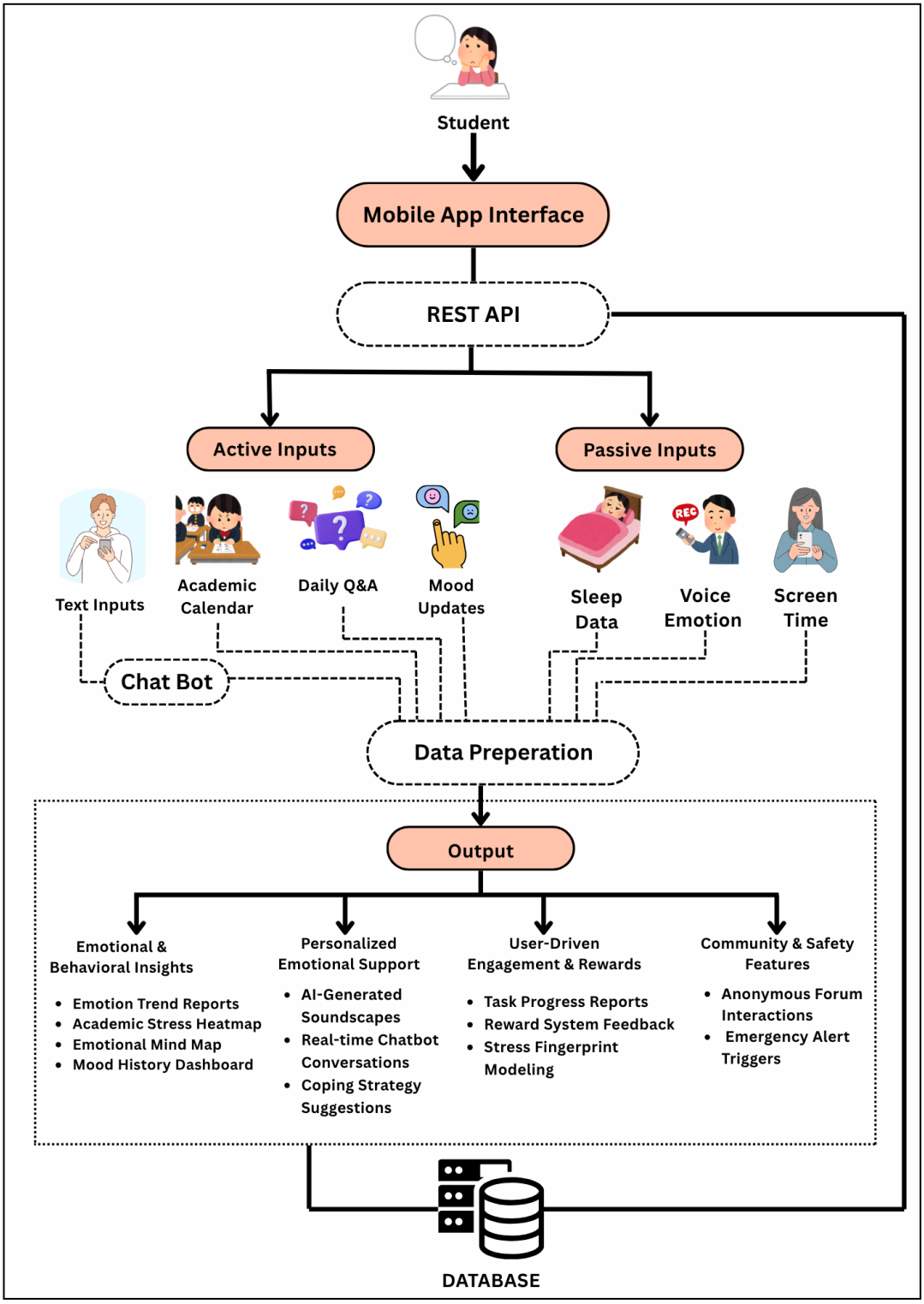

Figure 1 : System Diagram

The proposed self-monitoring and anonymous counselling-based digital companion follows a hybrid architectural approach that integrates both layered and microservice architectural styles. The mobile application communicates with the backend infrastructure through application programming interfaces (APIs), enabling seamless data exchange and functionalities. Each component of the system will be implemented as a microservice model, allowing for enhanced scalability and flexibility. This architectural style supports the rapid development and deployment of new features while ensuring the application handles the load, maintaining its performance and user experience.

The system depends on both active and passive user inputs, such as the onboarding questionnaire, mood logging, journaling, and chat access. Then the system generates a stress fingerprint for each user using clustering algorithms like K-Means or DBSCAN based on the answers to the onboarding questionnaire. Additionally, the system requires access to Google Fit to get sleep and screen activities. When users log into the app, they complete a daily questionnaire along with mood logs to assess their current situation. Based on this information, the system can suggest coping strategies if necessary. There is also a journaling feature that allows users to record their thoughts via voice, and after analyzing these voice journals, the system provides further coping strategy suggestions if needed, as well as recommendations for community interactions based on the journal entries. The community feature enables users to communicate with one another anonymously. Furthermore, the system includes a well-trained intelligent emotional assistant, with whom users can chat to receive solutions to their stress-related issues. Another component, known as the academic stress heatmap, helps users visualize their projected mood over the next three to five days based on their planned activities in a calendar, which will be discussed in more detail later.

3.2. Academic Stress Heatmap with Personalized Fingerprint Overlay

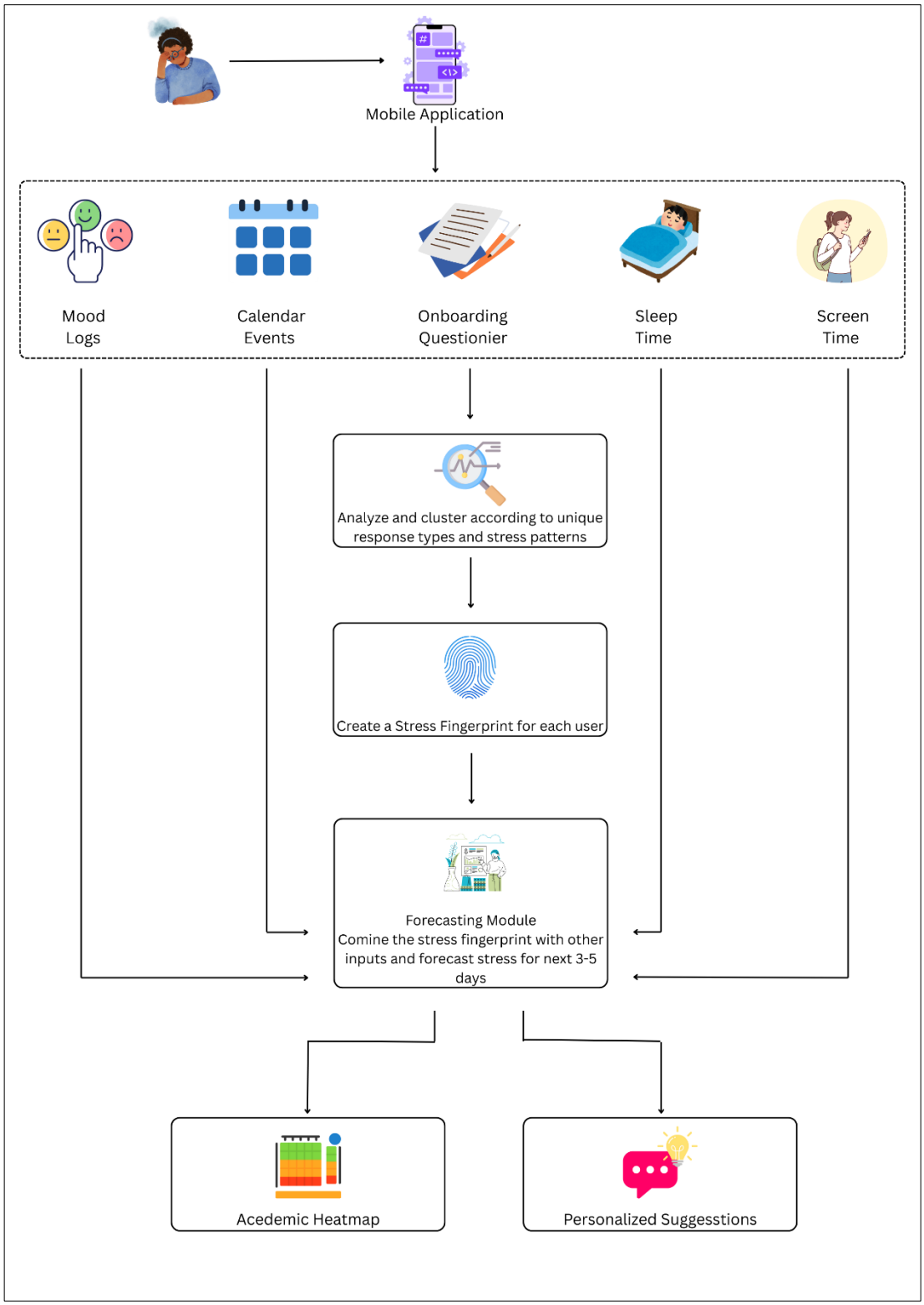

Figure 2 : Component Diagram

This proposed component can be divided into 5 different stages:

- **Data Collection**

This component requires both active and passive inputs. Active inputs like the onboarding questionnaire, daily mood logs, and calendar events will be captured via the user interface, while passive inputs like sleep and screen activities will be gathered using APIs. The onboarding questionnaire is created using validated psychological tools such as the COPE Inventory. [20], the Perceived Stress Scale (PSS) [21], or DASS-21 [22].

- **Generating the stress fingerprint using clustering algorithms**

The initial stress fingerprint will be generated using unsupervised clustering algorithms like K-Means or DBSCAN. At the initial stage, only the onboarding questionnaire will be used to identify the user's behavioral patterns to proceed with the fingerprint generation. While using the system for some time, other inputs like mood logs and journals can be integrated and update the fingerprint. For example, a student who constantly experiences high stress before deadlines will be identified as a deadline-sensitive while another student who is more affected by lack of sleep will be classified as sleep-reactive.

- **Generating the heatmap using predictive models**

The heatmap will be based on stress fingerprints, calendar events, and passive inputs. Time-series machine learning approaches like Long Short-Term Memory networks or Random Forest models will be trained on historical data combining mood, workload, and IoT inputs. These models will forecast stress levels three to five days ahead. These results will be displayed visually using a heatmap approach combined with a calendar view. The heatmap will display predicted stress with calendar events, color-coded from low (green) to high (red).

- **Personalized Suggestions**

Personalized suggestions will be provided based on forecasting results and stress patterns. For example, if the system predicts an increase in stress levels due to an

upcoming period filled with high workloads, it may suggest strategies such as breaking down large projects into smaller, more manageable tasks. This approach can help alleviate feelings of overwhelm and promote a sense of control. Additionally, if a disrupted sleep pattern is identified for a user who is sensitive to sleep changes, the system may suggest practical adjustments to their daily routine. This could include implementing a more consistent bedtime. Such targeted advice aims to help users navigate stress more effectively and improve their overall productivity and health.

3.3. Tools and Technologies

The chosen technologies for implementing the proposed component include React Native, Firebase, Python, and IoT integration via Google Fit, supported by visualization libraries and standard development tools.

- **React Native**

React Native will be used to develop the mobile application's user interface. Since this is a cross-platform framework, it reduces the development overhead and ensures uniformity across devices.

- **Firebase**

Firebase will serve as the backend infrastructure, enabling real-time data processing. [23] And cloud-based services. Additionally, Firebase offers a secure authentication process through Google, Apple, or email login. Its serverless architecture eliminated the need for custom server management, making Firebase the ideal choice for research-driven prototypes that may later scale.

- **Python**

Python will be used to train for all machine learning and predictive analytics tasks. Its ecosystem includes TenserFlow, PyTorch, and Scikit-learn, which provides flexibility for analysis and model development.

- **Google Fit**

Google Fit will be used to collect passive inputs. The APIs provided by Google Fit offer standardized access to health-related metrics such as sleep duration and screen time.

- **Visualization Frameworks**

For effective calendar heatmap visualization, Victory Native and React Native Chart Kit will be used to ensure clarity and usability for students

- **Supporting Tools**
    - Version Control : GitHub
    - Prototyping : Figma
    - API Testing : Postman
    - Model Experimentation : Jupyter Notebooks

4. WORK BREAKDOWN CHART

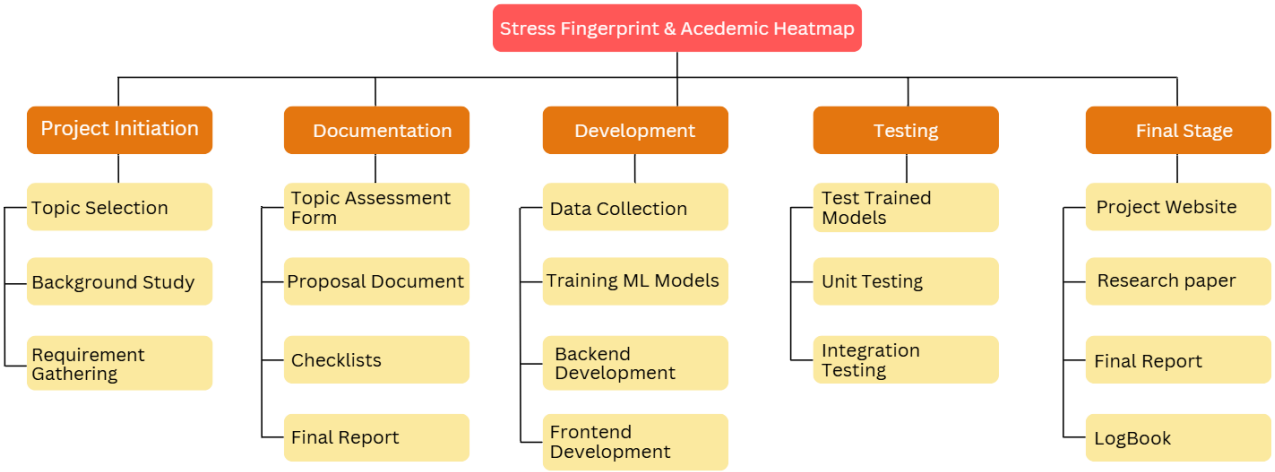

Figure 3 : Work Breakdown Structure

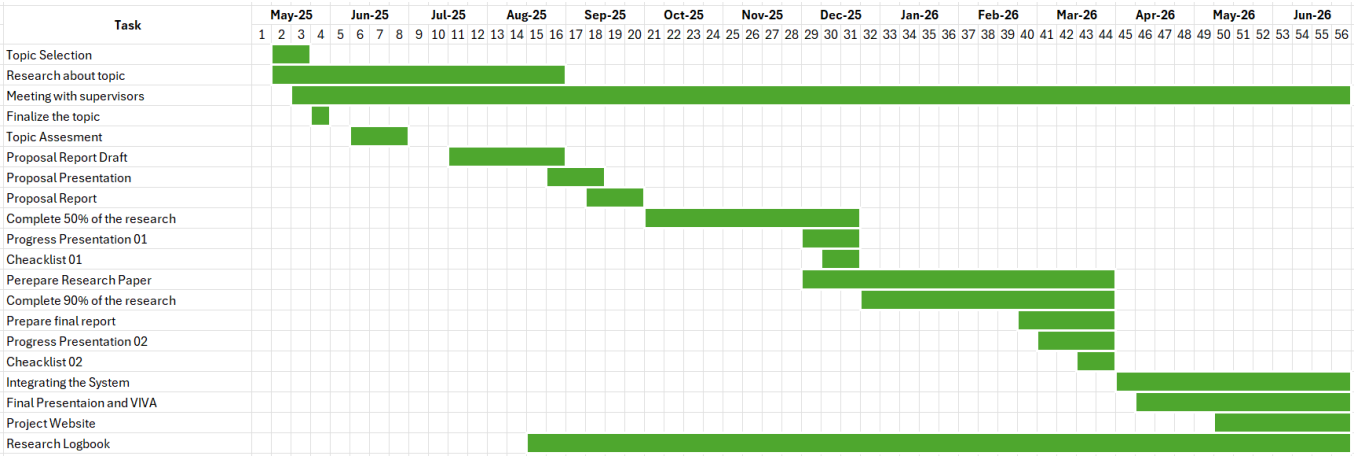

Figure 4 : Gantt Chart

5. PROJECT REQUIREMENTS

5.1. Functional Requirements

- **Daily mood and journal entry submission**

  Students can document their daily mood using short journal entries, numerical stress ratings, or emoji-based scales.

- **Generate stress fingerprint**

  The core innovation of the system lies in clustering students into behavioral profiles using unsupervised learning.

- **Integration with academic calendar and heatmap**

  A calendar-based heatmap will serve as the central visualization feature. Students will be able to see their past and predicted stress levels, color-coded by intensity. Academic deadlines and anticipated stress spikes will be overlaid on this interface, enabling students to foresee high-stress periods in advance. Suggest coping strategies

- **Anonymous community**

  To complement self-monitoring, the system will provide an anonymous chatroom feature where students can seek guidance by communicating with their peers.

5.2. Non-functional Requirements

- **Security**

  Due to the sensitive nature of mental health data, all communication between the client and server must be encrypted.

- **Reliability and Availability**

  The application must be available 24/7 to support students with varying schedules.

- **Compliance and Ethics**

  The system must adhere to ethical standards related to psychological research and data handling. Consent will be obtained before data collection.

- **Usability**

  The interface should be simple, visually engaging, and mobile-friendly, as most users will access the system using smartphones.

5.3. User requirements

- Easy onboarding process with simple questions to initialize the stress fingerprint.
- Smooth calendar integration for deadline tracking
- Clear visualization of stress levels on a calendar heatmap offering insights at glance.
- Timely and personalized reminders and coping suggestions
- Control over privacy to ensure the data remains secure and confidential.

5.4. System Requirements

- React-native for frontend development to ensure cross-platform availability
- Firebase for backend, authentication, and real-time storage.
- Python with libraries like TensorFlow for clustering and forecasting.
- Google Fit and HealthKit for passive IoT data.

5.5. Use Cases

Name	Creating Stress Fingerprint
ID	U001
Description	The system generates a personalized stress fingerprint for the user based on the onboarding questionnaire
Actors	User
Preconditions	User has installed the application and logged in
Steps	1. User completes the onboarding questionnaire. 2. Clustering algorithm analyzes data. 3. The system classifies the user into stress response types.
Outcome	User receives the initial stress fingerprint

Table 2 : Use case 1

Name	Stress Forecasting
ID	U002
Description	The system forecasts stress levels for the next 3-5 days
Actors	System
Preconditions	User has installed the application, logged in, completed the onboarding questionnaire, allowed access for wearable data collection, and inserted calendar events.
Steps	1. System collects recent mood logs, calendar events, and wearable data. 2. Data is preprocessed and input into the LSTM model. 3. System generates predicted stress values for upcoming days. 4. Predictions are stored and linked to visualization.
Outcome	Forecasted stress spikes are identified and prepared for display

Table 3 : Use case 2

Name	Heatmap Visualization
ID	U003
Description	The system presents stress predictions on a calendar heatmap with color coding
Actors	User
Preconditions	Stress fingerprints and predictions are available.
Steps	1. User navigates to the calendar in the app 2. System overlays academic deadlines and predicted stress levels 3. Calendar heatmap adjusts colors based on intensity and fingerprint type
Outcome	User visualizes upcoming stress spikes in a calendar

Table 4 : Use case 3

5.6. Test Cases

Module Name	Stress fingerprint creation		
Test Case ID	TC001		
Description	The system generates a stress fingerprint after onboarding		
Prerequisites	User has completed the onboarding questionnaire		
Test Steps	**Test Input**	**Expected Results**	**Comments**
Complete Onboarding questionnaire	Questionnaire answers	Data stored successfully	
Run clustering algorithm	Input dataset	System classifies fingerprint	
Save fingerprint to user profile	Classified fingerprint data	Fingerprint stored in firebase under user ID	

Table 5 : Test case 1

Module Name	Stress forecasting		
Test Case ID	TC002		
Description	The system forecasts stress spikes		
Prerequisites	At least 2 weeks of data		
Test Steps	**Test Input**	**Expected Results**	**Comments**
Load dataset	14+ days of logs and wearable data	Data preprocessed	
Run LSTM prediction	Forecast window = 5 days	System outputs predicted stress values	Check accuracy
Compare prediction vs actual	Real logs after 5 days	Accuracy > 75%	

Table 6 : Test case 2

Module Name	Heatmap Visualization		
Test Case ID	TC003		
Description	Display stress levels on a calendar heatmap		
Prerequisites	Predictions from the system		
Test Steps	**Test Input**	**Expected Results**	**Comments**
Open calendar view	User taps "Calendar"	Heatmap loads	
Check stress levels	Predicted values	Color coded heatmap (green-low, yellow-medium, red-high)	
Overlay academic deadlines	Exam on 20th September	Deadline displayed with stress zones	

Table 7 : Test case 3

5.7. Wireframes

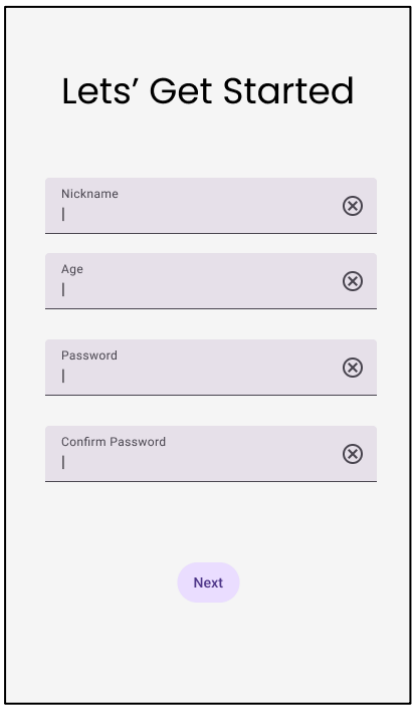

Figure 5 : Onboarding Screen 1

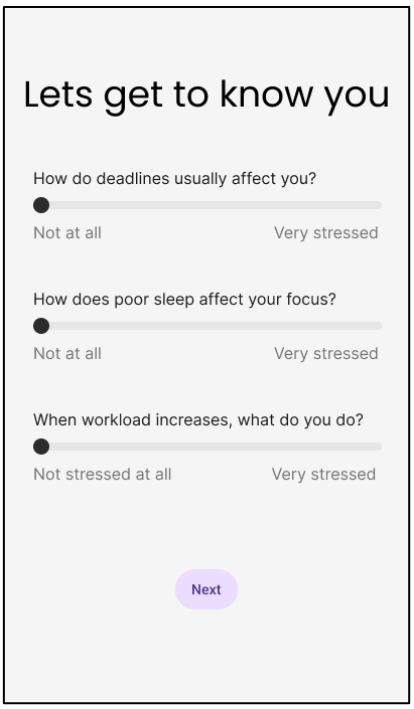

Figure 6 : Onboarding screen 2

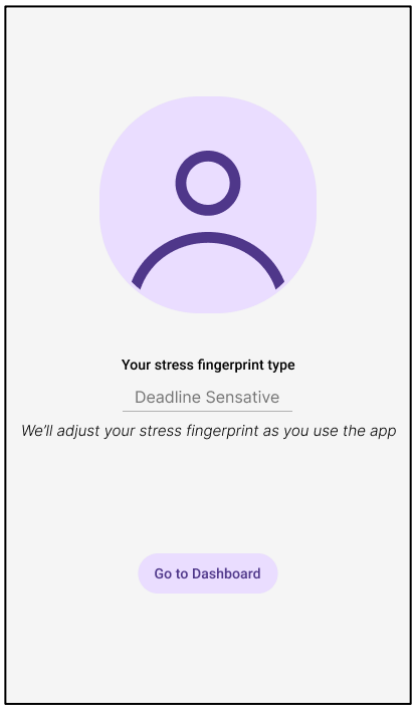

Figure 7 : Stress fingerprint

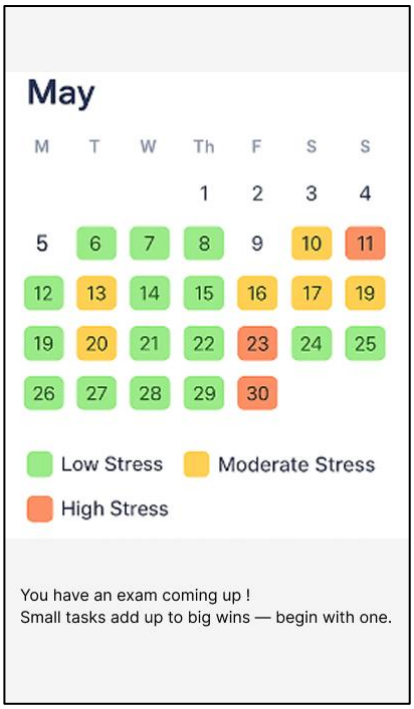

Figure 8 : Calander heatmap

6. BUDGET ANDGET JUSTIFICATIONS

Component	Allocated Budget	Justification
Software & Tools	Rs. 9000 per month	During development, the Firebase free tier may be sufficient. However, to manage larger data, a paid tier may be necessary.
Hosting Charges	Rs. 6600 per month	If necessary, the models should be hosted on AWS instead of Firebase.
Internet and Utilities	Rs. 10000	Required for development, online meetings
Stationary	Rs. 5000	Required for document printouts etc.

Table 8 : Budget Justification

The estimated total budget will be around Rs. 30,000.

Budget Justification

1. Software & Tools
   Most frameworks and libraries are free to use, significantly reducing costs. Only scaling Firebase or using cloud hosting will incur expenses.
2. Hardware
   Members' laptops and phones will be sufficient for development.
3. Miscellaneous
   Internet and printing costs are essential for smooth delivery.

7. COMMERCIALIZATION PLAN

Target Audience

- University Students
- Counsellors

Plans and Features

1. Free Plan
   - Access to mood tracking
   - Access to the academic calendar with a heatmap for predicted stress
   - Access to stress fingerprint analysis

   Price : Free

2. Premium Model
   - All features from the free plan
   - Personalized coping strategies to reduce stress in the upcoming days
   - Access to reports with past mood logs and predicted stress to share with counsellors or mentors

   Price : $10 per month

3. Institutional Access
   - Bulk subscriptions with discounts for universities.
   - Access up to 100 users
   - Full access to all premium features for all users
   - Access to Reports on anonymous insights about students' stress levels

   Price : $80 per month

REFERENCES

[1] World Health Organization, "Stress," 21 February 2023. [Online]. Available: https://www.who.int/news-room/questions-and-answers/item/stress. [Accessed 19 08 2025].

[2] R Beiter, R Nash, M McCrady, D Rhoades, M Linscomb, M Clarahan, S Sammut, "The prevalence and correlates of depression, anxiety, and stress in a sample of college students," *Journal of Affective Disorders,* vol. Volume 173, pp. Pages 90-96, 2015.

[3] World Health Organization, "Adolescent and young adult health," 26 November 2024. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/adolescents-health-risks-and-solutions. [Accessed 13 August 2025].

[4] P. Matthews, C. Rhodes-Maquaire, "Personalisation and Recommendation for Mental Health Apps: A Scoping Review," *Behaviour & Information Technology,* vol. 4, no. 10, pp. 2389-2404 , 2025.

[5] M. K. Moser, B. Resch, "An Individual-Oriented Algorithm for Stress Detection in Wearable Sensor Measurements," *IEEE Sensors Journal.,* 2023.

[6] K. Chawla, V. Sachdeva, "Domains of stress and coping strategies used by 1st year medical students," *National Journal of Physiology, Pharmacy and Pharmacology,* vol. 8, no. 3, pp. 366-369, 2018.

[7] N. Kumaraswamy, "Academic Stress, Anxiety and Depression among College Students- A Brief Review," *International Review of Social Sciences and Humanities,* vol. 5, no. 1, pp. 135-143, 2013.

[8] Y. Luo, I. Deznabi, A. Shaw, N. Simsiri, T. Rahman, M. Fiterau , "Dynamic clustering via branched deep learning enhances personalization of stress prediction from mobile sensor data," *Scientific Reports,* 2024.

[9] A. Sano, R. W. Picard, "Stress Recognition Using Wearable Sensors and Mobile Phones," in *Humaine Association Conference on Affective Computing and Intelligent Interaction*, Geneva, Switzerland, 2013.

[10] M. Alhasani, D. Mulchandani, O. Oyebode, N. Baghaei, R. Orji, "A Systematic and Comparative Review of Behavior Change Strategies in Stress Management Apps: Opportunities for Improvement," *Digital Public Health,* vol. 10, 2022.

[11] R. Wang, F. Chen, Z. Chen, T. Li, G. Harari, S. Tignor, X. Zhou, D. Ben-Zeev, A. T. Campbell, "StudentLife: assessing mental health, academic performance and behavioral trends of college students using smartphones," in *UbiComp '14: Proceedings of the 2014 ACM International Joint Conference on Pervasive and Ubiquitous Computing*, 2014.

[12] S. Gedam, S. Paul, "A Review on Mental Stress Detection Using Wearable Sensors and Machine Learning Techniques," *IEEE Access,* vol. 9, pp. 84045-84066, 2021.

[13] Z. Liang, O. Tatha, L. E. Andersen, "Developing mHealth App for Tracking Academic Stress and Physiological Reactions to Stress," in *2020 IEEE 2nd Global Conference on Life Sciences and Technologies (LifeTech)*, Kyoto, Japan, 2020.

[14] A. Shaw, N. Simsiri, I. Deznaby, M. Fiterau, T. Rahaman, "Personalized Student Stress Prediction with Deep Multitask Network," ArXiv, 2019.

[15] J. Torous, J. Nicholas, M. E. Larsen, J. Firth, H. Christensen, "Clinical review of user engagement with mental health smartphone apps: evidence, theory and improvements," *Evidence Based Mental Health,* vol. 21, no. 3, 2018.

[16] K. Wang, D. S. Varma, M. Prosperi, "A systematic review of the effectiveness of mobile apps for monitoring and management of mental health symptoms or disorders," *Journal of Psychiatric Research,* vol. 107, pp. 73-78, 2018.

[17] A. Bannore, T. Gore, A. Raut, K. Talele, "Mental stress detection using machine learning algorithm," in *International Conference on Electrical, Computer, Communications and Mechatronics Engineering (ICECCME)*, Mauritius, 2021 .

[18] M. Danieli, T. Ciulli, S. M. Mousavi, G. Riccardi, "A Conversational Artificial Intelligence Agent for a Mental Health Care App: Evaluation Study of Its Participatory Design," *JMIR Form Res,* vol. 5, no. 12, 2021.

[19] B. Inkste, S. Sarda, V. Subramanian, "An Empathy-Driven, Conversational Artificial Intelligence Agent (Wysa) for Digital Mental Well-Being: Real-World Data Evaluation Mixed-Methods Study," *JMIR Mhealth Uhealth,* vol. 6, no. 11, 2018.

[20] C. S. Carver, "COPE (complete version)," [Online]. Available: https://www.psy.miami.edu/faculty/ccarver/copefull.html. [Accessed 2025 08 20].

[21] K. M. Harris, A. E. Gaffey, J. E. Schwartz, D. S. Krantz, M. M. Burg, "The Perceived Stress Scale as a Measure of Stress: Decomposing Score Variance in Longitudinal Behavioral Medicine Studies," *Ann Behav Med,* vol. 10, p. 846–854, 2023.

[22] S. H. Lovibond, P. F. Lovibond, "Dass21 questionnaire," in *Manual for the Depression Anxiety & Stress Scales*, Sydney, 1995.

[23] A. Efendi, M. I. Ammarullah, I. G. T. Isa, M. P. Sari, J. N. Izza, Y. S. Nugroho, H. Nasrullah, D. Alfian, "IoT-Based Elderly Health Monitoring System Using Firebase Cloud Computing," *Health Science Reports,* vol. 8, no. 3, 2025.